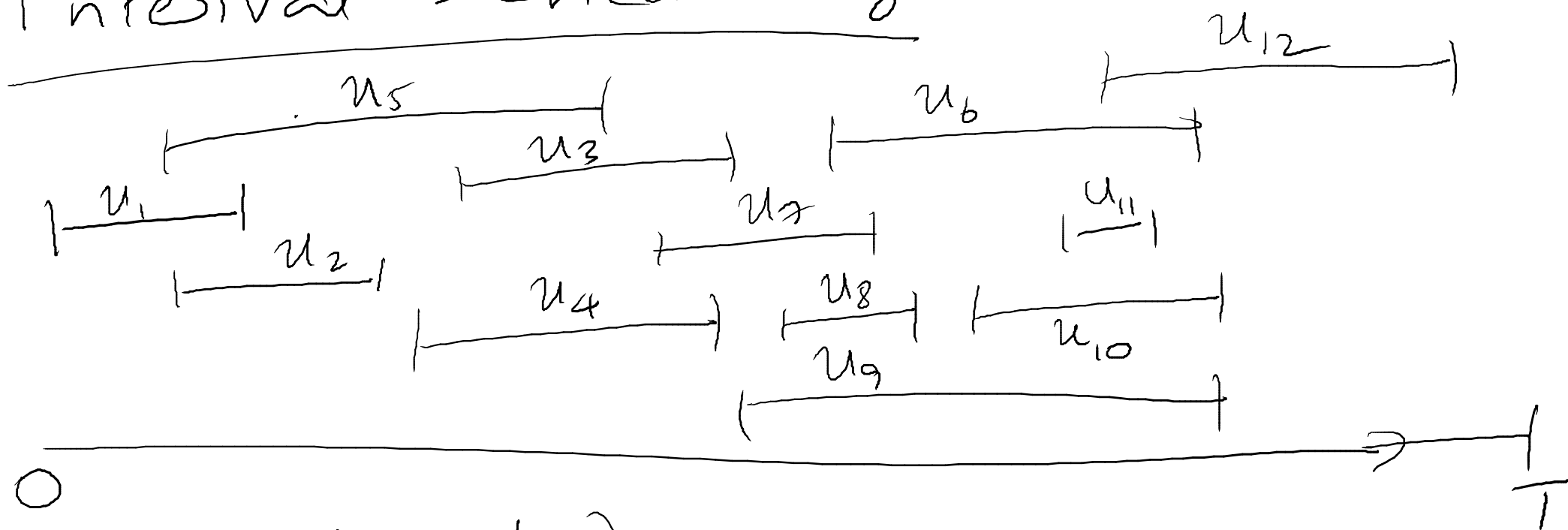


Interval Scheduling (Weighted)



$$u_i = (s_i, t_i) \quad v_i$$

Goal! Maximize the total value jobs that can be completed.

Dynamic Prog. Approach

If $u_{12} \in OPT$,

- remove all conflict jobs

- find OPT soln for $u_1 \dots u_n$

If $u_{12} \notin OPT$

($u_{n+1} \dots u_{12}$ conflicts with u_{12})

- find OPT for $u_1 \dots u_{11}$

$$\text{val OPT}(u_1, \dots, u_n)$$

$$= \max \begin{cases} v_{12} + \text{val OPT}(u_1, \dots, u_n) \\ \text{val OPT}(u_1, \dots, u_n) \end{cases}$$

Memoization

$O(n^2)$ sdn

Read: Section 6.1, 6.2 in
Klienbergr & Tardos

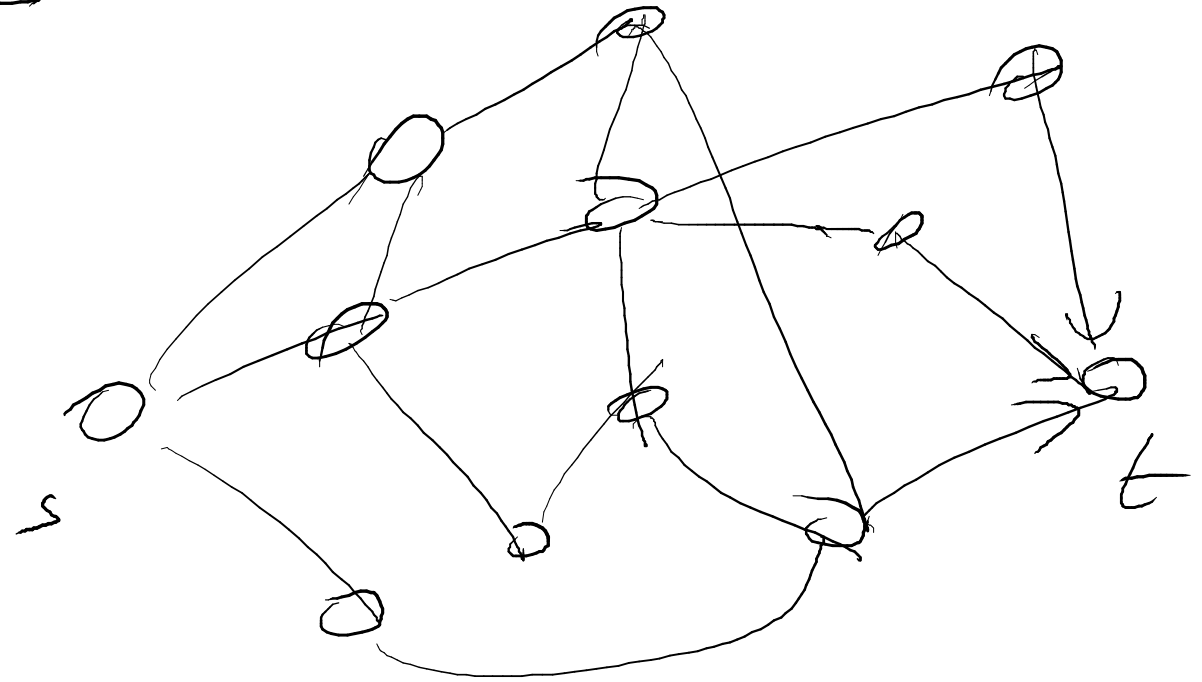
Shortest Paths in DAG

— Topo Sort

u_1

u_2

$s \ u_2 \ u_3$



$u_n \ t = u_n$

$$\text{OPT}(u_1 \dots u_n) = \min_{\substack{\text{pred}(t) \\ u'}} l(u' \rightarrow t) + \text{OPT}(1 \dots u')$$

Read: Section 6.1 in
Dasgupta, Papadimitriou, Vazirani

Read

- Section 6.1, 6.2 in Klienberg, Tardos
- Section 6.1 in Dasgupta, Papadimitriou, Vazirani

Solve

- Exercise 6.2, 6.5, 6.6 in Dasgupta, Papadimiriu, Vazirani
- Chapter 6, Exercise 1 in Klienberg, Tardos